Student Activity Sheet

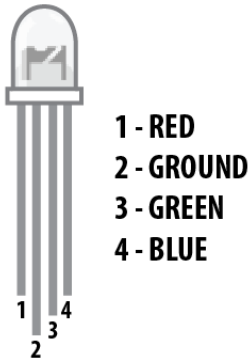
Activity 1H: Create any color with RGB LEDs

Purpose: You know what's even more fun than a blinking LED? Changing colors with one LED. RGB, or red-green-blue, LEDs have three different color-emitting diodes that can be combined to create all sorts of colors. In this circuit, you'll learn how to use an RGB LED to create unique color combinations. Depending on how bright each diode is, nearly any color is possible!

Materials: Computer, breadboard, Arduino Uno, 1x LED - RGB Common Cathode, 3x 220Ω Resistors, 5x Jumper Wires

Note: You must use "cathode" LEDs to work properly with the Arduino. They have a common negative cathode. If you accidentally use an anode LEDs it will not damage the board, but it must be wired and coded differently.

New Ideas and Materials: Up until this point, we have used LEDs that are only a single color. We could only change the color by changing the LED. Wouldn't it be cool if we could choose any color we wanted? What about teal, purple, orange, or even more??? Introducing our new friend, the RGB LED. An RGB LED is really three smalls LEDs next to each other. A Red one, a Green one, and a Blue one. (Hence, why it is called RGB). It turns out that you can make any color by mixing these three light colors together.

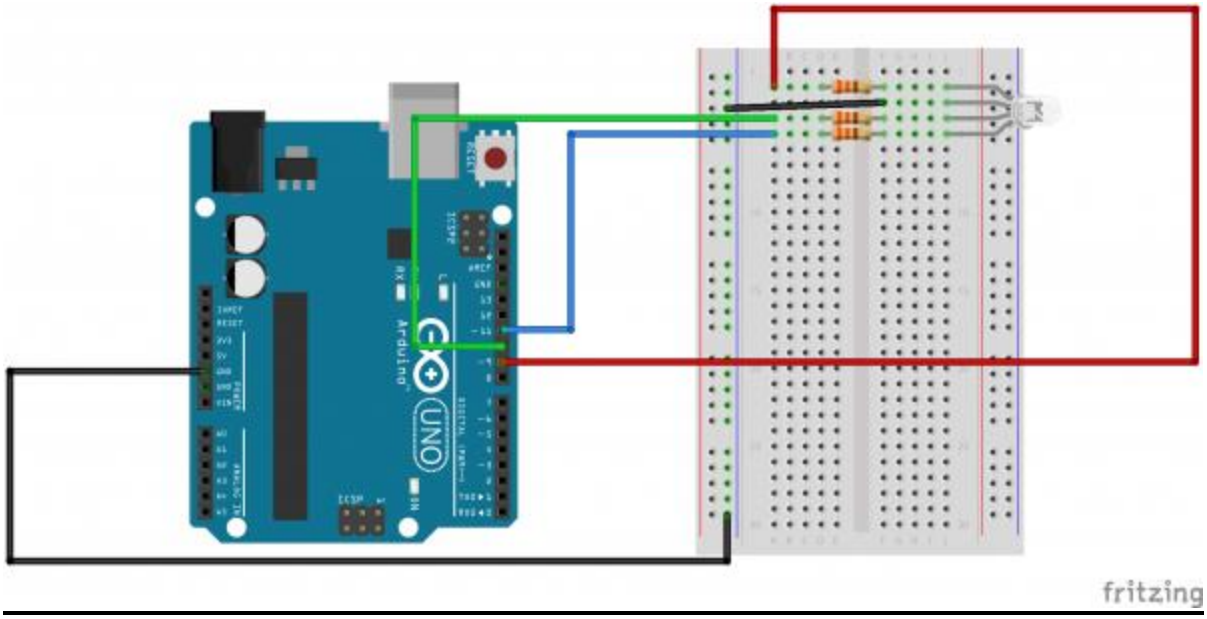


Note: When wiring the RGB, each colored pin still needs a current-limiting resistor in-line with the I/O pin that you plan to use to control it, as with any standard LED.

Circuit: Ready to start hooking everything up? Check out the Fritzing diagram and hookup table below, to see how everything is connected.

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Pay special attention to the component's markings indicating how to place it on the breadboard. Polarized components can only be connected to a circuit in one direction.

Code

The RGB LED can emit different colors by mixing the 3 basic colors red, green and blue. So, it consists of 3 separate LEDs red, green and blue packed in a single case. That's why it has 4 leads, one lead for each of the 3 colors and one common cathode or anode depending of the RGB LED type. In this tutorial I will be using a common **cathode** LED.

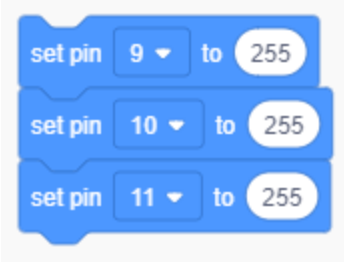
1. You can create your code In the TinkerCAD website

2. To create colors, we need to turn each LED on and off separately. If we just turned the LEDs on or off, we could only create 8 colors. So, we will use PWM (Pulse Width Modulation) value from 0 (all the way off) to 255 (all the way on) once again to allow us to create any color in the rainbow.

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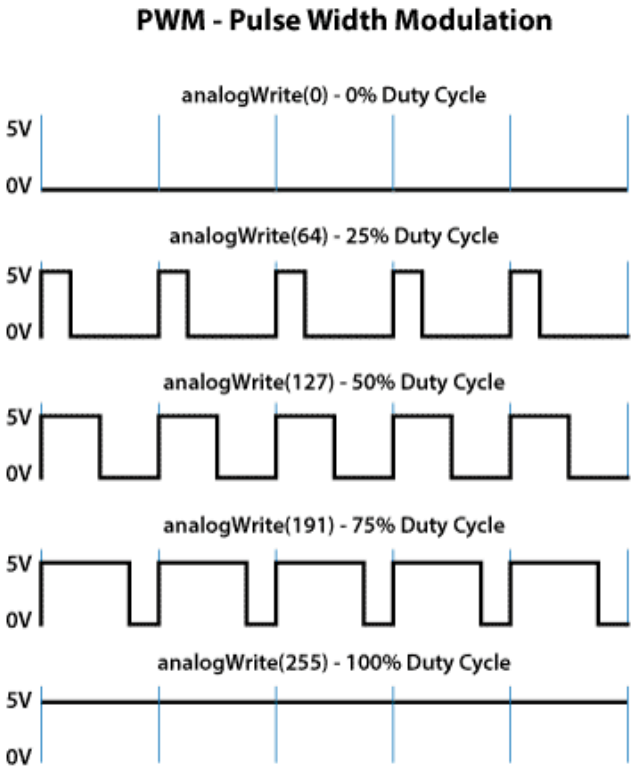
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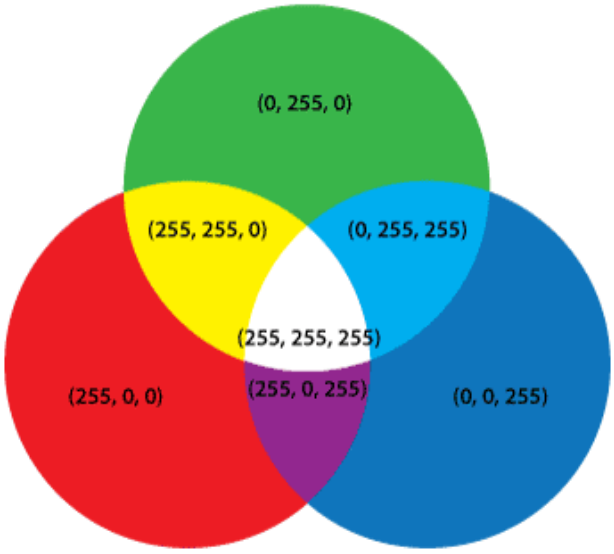
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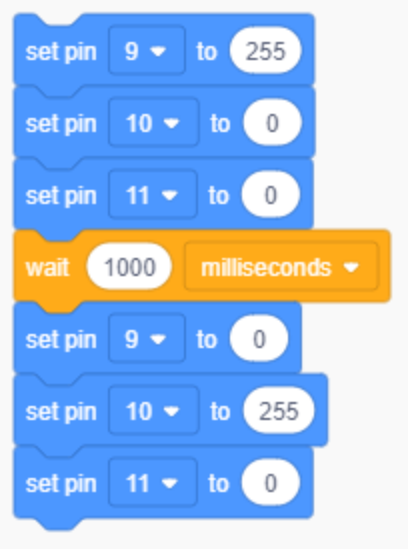
Upload this code to your Arduino and try changing the values. The ones shown simply give you white. **You can adjust the amount of red, green or blue by using the 0 to 255 scale. 0,0,0 will result in black.**

3. Now create code to alternate between red, green, blue, yellow, magenta, cyan, white and black. Hold each color for 1 second.

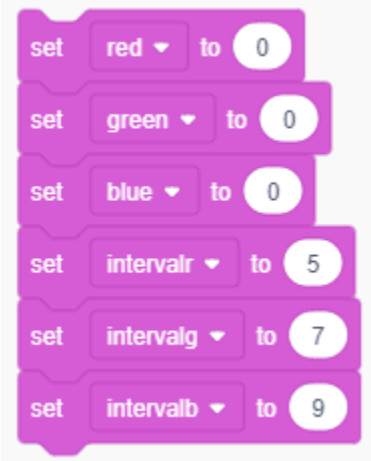




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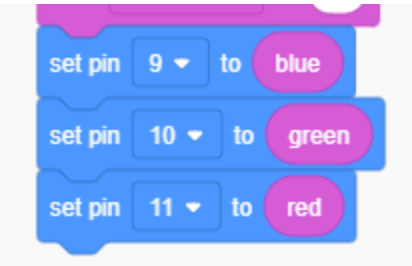


4. Next, attempt to change the value of the PWM setting with variables. We first create the variables similarly to Activity 1F. These will be integer variables red, green, blue and three more for changing the brightness or PWM length.



5. The set statement is added to tell the led in pin 9, 10 and 11 how bright it is. Rather than giving a constant high or low value we use the brightness variable.

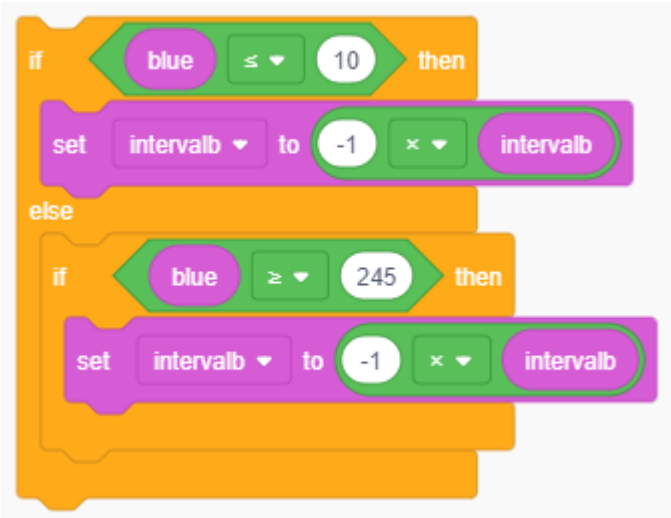
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6. An increment counter is usually used to increment the LED brightness slowly. The set statement is used to add or subtract a value of 5 from the brightness.

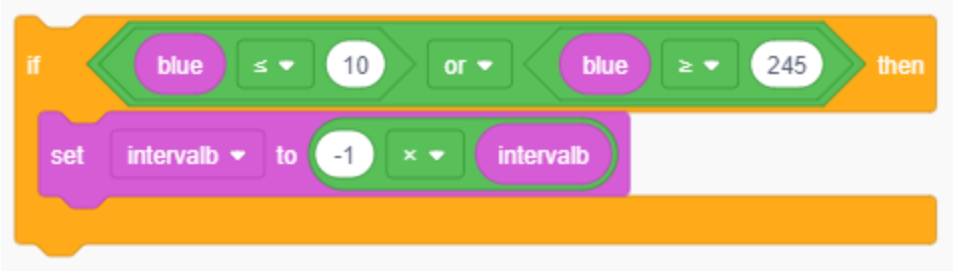


7. If brightness is at either extreme of its value (either 0 or 255), then variable “intervalb” is changed to its negative. In other words, if intervalb is 5, then it is set to -5. If it's -5, then it's set to 5. The next time through the loop, this change causes brightness to change direction as well. The “if” statement is used to set the intervalb to positive or negative 5 if the value gets to 255 to then decrease and once at zero to increase brightness.



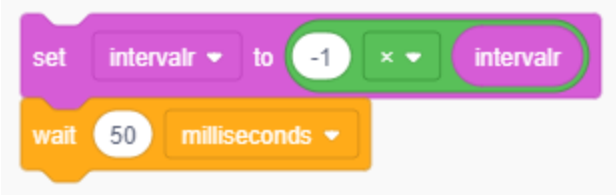
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Note that you can copy entire if statements and their blocks. Just right click and duplicate. This will save time when setting up red and green.

8. The “delay” at the end controls how fast the fading occurs. It measures in milliseconds. Try changing the value of the delay and see how it changes the fading effect.



*****SAVE your sketch to your Folder that you created*****

Save your sketch as “RGBrainbow.”

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9. Copy and paste the Arduino Code from the " TinkerCAD website" Hook up your Arduino to your computer using the USB cable.

```
int red = 0;

int green = 0;

int blue = 0;

int intervalr = 0;

int intervalg = 0;

int intervalb = 0;

void setup()
{
  pinMode(9, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(11, OUTPUT);
}

void loop()
{
  red = 0;
  green = 0;
  blue = 0;
  intervalr = 5;
  intervalg = 7;
  intervalb = 9;
  analogWrite(9, blue);
  analogWrite(10, green);
  analogWrite(11, red);
  blue = intervalb;
  green = intervalg;
  red = intervalr;
  if (blue <= 10) {
    intervalb = (-1 * intervalb);
  } else {
    if (blue >= 245) {
      intervalb = (-1 * intervalb);
    }
  }
  intervalr = (-1 * intervalr);
  delay(10); // Delay a little bit to improve simulation performance
}
```

Observe your LED and complete your worksheet titled Assignment 1H: Create any color with RGB LEDs

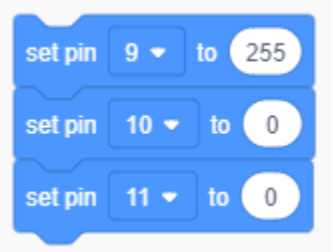
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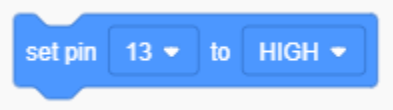
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Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the led, is the long positive arm connected by wires to the pins?
    c. It may be wise to test the each led by turning them on one at a time. Open a new window and create this test code. The blocks should look like below. Then change the values to turn each one on.



    d. LED still not working? Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when you try this test code?



    e. Next check the wires in the circuit. This can be done by connecting the LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.

Assignment 1H: Create any color with RGB LEDs

1. Look carefully at the pins we used for the analog code block. You may have noticed only certain pins were in the dropdown menu. What do you notice marked on the UNO board that designates the analog output pins?

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2. **THE CODE.** Code is a set of instructions used to communicate to the Arduino.
    a. Describe in detail what happened to the colors in section 1, as you changed the analog numbers between 0 and 255?

    b. Initially, we set the “interval” variables to 5, 7 and 9. Why not all just to 5?



    c. You may have also noticed the If statements use 10 and 245 for the turnaround point. Why do you think it is? What happens if you use 0 and 255?

    d. What happens when you alter the delay at the end from 50 to 100 or 10?

Extension Challenges:

3. Change your code so that your LED flashes color at 5 second intervals. Explain how you changed your code to achieve this.

4. Change your code and add a second RGB LED on another pin changing color at a different rate. You may have to introduce additional variables to achieve this.

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5. Connect a button switch to your circuit Code your Arduino to start the colors changing when the button is pressed.

CONCLUSION:

6. List two real life applications that this type of RGB LEDs.

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